

West Bengal Council of Higher Secondary Education
EXPLAINED ANSWER PAPER

(Paper 2A — Questions with Answers)

Modern Computer Application (COMA)

Class XII – Theory

Unit – 2 : Data Warehousing and Data Mining

Full Marks : 70

Detailed answer with
explanation for each question

In this answer paper every question is restated and followed by a detailed model answer with explanation. Examples and comparison tables are included so that students can understand each topic deeply.

**GROUP – A (Multiple Choice Type Questions) [21 × 1
= 21]**

Q1. A collection of subject-oriented, integrated, time-variant and non-volatile data is called a — (a) Database (b) Data Warehouse (c) Data File (d) Spreadsheet [1]

Answer & Explanation:

Correct answer: (b) **Data Warehouse.**

A data warehouse has four key features — subject-oriented, integrated, time-variant and non-volatile. It brings together historical data from many sources to support analysis and decision making. A normal database is mainly for daily transactions, and a spreadsheet is for small data.

Q2. OLAP stands for — (a) Online Analytical Processing (b) Online Access Protocol (c) Offline Analytical Program (d) Online Application Process [1]

Answer & Explanation:

Correct answer: (a) **Online Analytical Processing.**

OLAP quickly analyses multidimensional data to support complex queries and decision making; it is mainly used with a data warehouse. The other options are not the real full form.

Q3. OLTP stands for — (a) Online Transaction Processing (b) Online Testing Program (c) Offline Transaction Protocol (d) Online Table Processing [1]

Answer & Explanation:

Correct answer: (a) **Online Transaction Processing**.

OLTP quickly handles day-to-day business transactions (e.g., banking, ticket booking) with many small read/write operations. It is the opposite of OLAP, which is analysis-oriented.

Q4. The multidimensional structure of data in a data warehouse is called a — (a) Data Cube (b) Data Row (c) Data Key (d) Data Link [1]

Answer & Explanation:

Correct answer: (a) **Data Cube**.

A Data Cube arranges data along several dimensions (e.g., time, place, product) to allow multidimensional analysis. It is the basis of OLAP operations. Data Row, Key or Link are not multidimensional structures.

Q5. Which of the following is NOT a feature of a data warehouse? (a) Subject-oriented (b) Integrated (c) Volatile (d) Time-variant [1]

Answer & Explanation:

Correct answer: (c) **Volatile**.

A data warehouse is actually **non-volatile** — once stored, data is not easily changed or deleted but kept permanently for analysis. So “Volatile” is not a feature. The other three are correct features.

Q6. The process of discovering unknown and meaningful patterns from a large data store is called — (a) Data Mining (b) Data Entry (c) Data Backup (d) Data Locking [1]

Answer & Explanation:

Correct answer: (a) **Data Mining**.

Data mining discovers previously unknown, meaningful patterns and relationships from a large database or data warehouse to help with prediction and decision making. Data entry inputs data, backup stores it, and locking is security-related.

Q7. One of the first steps of the data mining process is — (a) Data Cleaning (b) Data Deletion (c) Data Hiding (d) Data Encryption [1]

Answer & Explanation:

Correct answer: (a) **Data Cleaning**.

Data mining begins with Data Cleaning, where noise, missing values and inconsistencies are removed so that the analysis is accurate. Deletion, hiding and encryption are not data mining steps.

Q8. The step that removes noise and missing values is called — (a) Data Transformation (b) Data Cleaning (c) Data Mining (d) Data Selection [1]

Answer & Explanation:

Correct answer: (b) **Data Cleaning**.

In the Data Cleaning step, noise, errors and missing values are corrected or filled in. Data Transformation converts data into a suitable form, and Data Selection chooses relevant data.

Q9. The step of combining data from multiple sources is called — (a) Data Integration (b) Data Reduction (c) Data Cleaning (d) Data Coding [1]

Answer & Explanation:

Correct answer: (a) **Data Integration**.

Data Integration combines data from multiple sources (several databases, files) into a consistent form. Data Reduction reduces the volume of data, and Cleaning removes errors.

Q10. The step of converting data into a form suitable for analysis is called — (a) Data Transformation (b) Data Deletion (c) Data Locking (d) Data Backup [1]

Answer & Explanation:

Correct answer: (a) **Data Transformation**.

In Data Transformation, data is converted through normalization, aggregation or encoding into a form suitable for mining. The other options are not related to transformation.

Q11. Normalization is a strategy of which step? (a) Data Cleaning (b) Data Transformation (c) Data Mining (d) Data Backup [1]

Answer & Explanation:

Correct answer: (b) **Data Transformation**.

Normalization is a strategy of Data Transformation, where values are scaled to a fixed range (e.g., 0–1) so that all attributes get equal importance. It is neither Cleaning nor Backup.

Q12. OLAP is mainly used with a — (a) Data Warehouse (b) Web Browser (c) Printer (d) Keyboard [1]

Answer & Explanation:

Correct answer: (a) **Data Warehouse**.

OLAP is mainly used to analyse the multidimensional data of a data warehouse — such as slice, dice and drill-down operations. It has no direct link with a browser, printer or keyboard.

Q13. Day-to-day business transactions are handled by — (a) OLTP (b) OLAP (c) Data Mining (d) Data Cube [1]

Answer & Explanation:

Correct answer: (a) **OLTP**.

OLTP (Online Transaction Processing) handles day-to-day transactions such as banking, sales and booking. OLAP is for analysis, Data Cube is a multidimensional structure, and Data Mining finds patterns.

Q14. Analysing data collected in time order is called — (a) Time Series Mining (b) Text Mining (c) Web Mining (d) Key Mining [1]

Answer & Explanation:

Correct answer: (a) **Time Series Mining**.

Time series data is a sequence of values collected in time order (e.g., daily temperature or share price); its analysis reveals trends and enables prediction. Text and Web mining are different types.

Q15. Data mining on data such as DNA or protein is called — (a) Biological Data Mining (b) Audio Mining (c) Visual Mining (d) Symbolic Mining [1]

Answer & Explanation:

Correct answer: (a) **Biological Data Mining.**

Biological Data Mining analyses biological data (DNA sequences, protein structures, genes) to help in disease study, heredity and drug research. It is an important application of bioinformatics.

Q16. Extracting patterns from images or video is called — (a) Visual Data Mining (b) Text Mining (c) Sequence Mining (d) Statistical Mining [1]

Answer & Explanation:

Correct answer: (a) **Visual Data Mining.**

Visual Data Mining extracts meaningful patterns and information from images and video (and also presents mining results visually). Text mining works with written text.

Q17. Analysing ordered events is called — (a) Sequence Data Mining (b) Audio Mining (c) Cube Mining (d) Row Mining [1]

Answer & Explanation:

Correct answer: (a) **Sequence Data Mining.**

Sequence Data Mining analyses the order of successive events or transactions (e.g., a customer first buys a mobile, then a cover). It is useful in recommendation systems.

Q18. Analysing data using statistical methods is called — (a) Statistical Data Mining (b) Visual Mining (c) Web Mining (d) Key Mining [1]

Answer & Explanation:

Correct answer: (a) **Statistical Data Mining.**

Statistical Data Mining uses statistical methods such as mean, deviation, regression and probability to extract relationships and patterns from data.

Q19. Which of the following is a real application of data mining? (a) Market Basket Analysis (b) File Printing (c) Data Typing (d) Screen Locking [1]

Answer & Explanation:

Correct answer: (a) **Market Basket Analysis.**

Market Basket Analysis finds the pattern of products bought together (e.g., “bread is often bought with butter”) — a famous application of data mining. Printing or typing are not applications.

Q20. Which is an example of data that can be mined? (a) Relational and Transactional data (b) only empty files (c) only broken disks (d) only empty tables [1]

Answer & Explanation:

Correct answer: (a) **Relational and Transactional data.**

Data mining can be applied to relational databases, transactional databases, data warehouses, text, web and multimedia data. Empty files or empty tables have no data to mine.

Q21. A future trend in data mining is — (a) Big Data and AI-based mining (b) paper-based accounting (c) manual calculation (d) using a typewriter[1]

Answer & Explanation:

Correct answer: (a) **Big Data and AI-based mining.**

Future trends include using AI and machine learning on huge volumes of Big Data for fast, automatic mining, along with cloud-based and real-time analysis. Paper-and-pencil methods are outdated.

GROUP – B (Short Answer Type Questions) [14 × 1 = 14]

Q1. What is a Data Warehouse? [1]

Answer & Explanation:

A **Data Warehouse** is a central store of subject-oriented, integrated, time-variant and non-volatile data that brings together historical data from many sources to support analysis and managerial decision making.

Q2. Write any two features of a Data Warehouse. [1]

Answer & Explanation:

(1) **Subject-oriented** — organised around a specific subject (e.g., sales, customers).
(2) **Non-volatile** — once stored, data is not easily changed or deleted. (Extra: Integrated and Time-variant.)

Q3. What is meant by Data Warehouse Modelling? [1]

Answer & Explanation:

Data Warehouse Modelling is the design process of arranging data for analysis, using fact and dimension tables to build structures such as the **Star Schema** or **Snowflake Schema**.

Q4. What is a Data Cube? [1]

Answer & Explanation:

A **Data Cube** is a multidimensional structure of data in which data is arranged along several dimensions (e.g., time, place, product) so that fast analysis (OLAP) can be performed.

Q5. What is OLAP? [1]

Answer & Explanation:

OLAP (Online Analytical Processing) is a technology that quickly analyses the multidimensional data of a data warehouse to support complex queries, summaries and decision making (slice, dice, drill-down).

Q6. Write any one difference between OLTP and OLAP. [1]

Answer & Explanation:

OLTP handles day-to-day transactions (current data, many small read/write operations), whereas **OLAP** analyses historical data to support decision making (complex queries, mostly read).

Q7. What is Data Mining? [1]

Answer & Explanation:

Data Mining is the process of discovering previously unknown, meaningful patterns, relationships and knowledge from a large data store. It helps in prediction and decision

making.

Q8. What kinds of data can be mined? Give two examples. [1]

Answer & Explanation:

Data mining can be applied to relational databases, transactional databases, data warehouses, text, web and multimedia data. Examples: (1) a shop's transactional sales data, (2) a website's log data.

Q9. What is Data Cleaning? [1]

Answer & Explanation:

Data Cleaning is the initial step of data mining in which noise, errors, inconsistencies and missing values are corrected or filled in, so that the analysis is accurate and reliable.

Q10. What is Data Transformation? [1]

Answer & Explanation:

Data Transformation is the step in which data is converted into a form suitable for mining — for example, normalization, aggregation, smoothing and generalization.

Q11. What is Time Series data? [1]

Answer & Explanation:

Time Series data is a set of values collected at equal intervals in time order (e.g., daily temperature, monthly sales or share prices), used to analyse trends and make predictions.

Q12. What is Sequence Data Mining? [1]

Answer & Explanation:

Sequence Data Mining is the process of finding patterns within the order of successive events or transactions (e.g., a customer's sequential purchases), which helps in recommendation and behaviour analysis.

Q13. Give one example of Biological Data Mining. [1]

Answer & Explanation:

Analysing a DNA sequence to identify genes related to a disease is an example of **Biological Data Mining**; it is used in drug discovery and heredity research.

Q14. Mention any two real applications of Data Mining. [1]

Answer & Explanation:

(1) **Market Basket Analysis** — finding relationships between products bought together. (2) **Fraud Detection** — detecting fraud in banking. (Extra: customer segmentation, disease diagnosis.)

GROUP – C (Broad / Descriptive Type Questions) [5 × 7 = 35]

Q1. (a) What is a Data Warehouse? Explain its main features. (b) Briefly explain Data Warehouse Modelling. [4+3]

Answer & Explanation:

(a) Data Warehouse & Features: a central store of integrated historical data from various sources, used for analysis and decision making. Its four main features are —

- **Subject-oriented** — organised around a subject (sales, customers).
- **Integrated** — data from various sources unified into one form.
- **Time-variant** — holds historical data over long periods.
- **Non-volatile** — data is permanent, not easily changed/deleted.

(b) Data Warehouse Modelling: the design of arranging data for analysis using a **fact table** (numeric measures) and **dimension tables** (descriptive data). The two main schemas are —

- **Star Schema** — one central fact table surrounded by dimension tables.
- **Snowflake Schema** — dimensions further split into normalized sub-tables.

Q2. (a) Explain what a Data Cube is, with an example. (b) Write the differences between OLAP and OLTP. [3+4]

Answer & Explanation:

(a) Data Cube: a multidimensional structure of data along several dimensions, allowing fast analysis. Example: if sales data is arranged by three dimensions — *time*, *place*, *product* — then “sales of mobiles in Kolkata in 2024” is found quickly in one cell. Operations like slice, dice, drill-down and roll-up can be performed on it.

(b) OLAP vs OLTP:

OLTP	OLAP
Handles day-to-day transactions	Analysis and decision making
Many small read/write operations	Complex queries, mostly read
Current data	Historical and summarised data
Example: banking, booking	Example: sales analysis, reports

Q3. (a) What is Data Mining? Explain what kinds of data and patterns can be mined. (b) Briefly discuss Data Warehouse Implementation. [4+3]

Answer & Explanation:

(a) **Data Mining, data & patterns:** the process of discovering unknown, meaningful patterns from a large data store.

- **Data that can be mined** — relational, transactional, data warehouse, text, web and multimedia data.
- **Patterns that can be mined** — Association (co-occurrence), Classification, Clustering, Prediction and Outlier detection.

(b) **Data Warehouse Implementation:** the steps to build a data warehouse —

- **ETL** — Extract, Transform and Load data from various sources.
- **Storage** — store in fact and dimension tables (Star/Snowflake schema).
- **OLAP server** — provides multidimensional analysis.
- **Front-end tools** — for query, reporting and visualization.

Q4. (a) Explain the main steps of the Data Mining process. (b) Write the importance of Data Cleaning. [4+3]

Answer & Explanation:

(a) **Steps of the Data Mining process:**

1. **Data Cleaning** — remove noise and missing values.
2. **Data Integration** — combine data from multiple sources.
3. **Data Selection** — select relevant data.
4. **Data Transformation** — convert into a suitable form.
5. **Data Mining** — extract patterns (the core step).
6. **Pattern Evaluation** — verify meaningful patterns.
7. **Knowledge Presentation** — present results as charts/reports.

(b) **Importance of Data Cleaning:** dirty or wrong data leads to wrong results (“garbage in, garbage out”). Data cleaning removes noise, errors and missing values, improving data quality so that the mining results are accurate, reliable and useful for decision making.

Q5. (a) What is Data Transformation? (b) Explain the various strategies of Data Transformation. [3+4]

Answer & Explanation:

(a) **Data Transformation:** the step of converting data into a suitable and consistent form for mining, so that the analysis becomes easy and accurate.

(b) **Strategies:**

- **Normalization** — scale values to a fixed range (e.g., 0–1).

- **Aggregation** — summarise data (e.g., daily to monthly sales).
- **Smoothing** — remove noise to make data smooth.
- **Generalization** — convert low-level data into higher-level concepts (e.g., age → young/old).
- **Attribute construction** — create new attributes from existing ones.

Q6. (a) Explain Sequence Data, Time Series and Statistical Data Mining. (b) Discuss Biological Data Mining with an example. [4+3]

Answer & Explanation:

(a) Three types of Mining:

- **Sequence Data Mining** — finds patterns in the order of successive events (e.g., a customer's sequential purchases).
- **Time Series Mining** — analyses data collected in time order (e.g., share prices) for trends and prediction.
- **Statistical Data Mining** — finds relationships using statistical methods such as mean, regression and probability.

(b) Biological Data Mining: analysing biological data such as DNA, RNA, proteins and genes to extract meaningful patterns. Example — analysing a DNA sequence to identify disease-related genes, useful in drug discovery and heredity research. It is an important field of bioinformatics.

Q7. (a) Explain Visual and Audio Data Mining. (b) Discuss the various applications and future trends of Data Mining. [4+3]

Answer & Explanation:

(a) Visual & Audio Data Mining:

- **Visual Data Mining** — extracting patterns from images, figures and video, and presenting mining results in visual form (graphs, charts) for easy understanding.
- **Audio Data Mining** — analysing sound and audio signals (e.g., voice, music) to extract patterns; sometimes data is also presented in audio form.

(b) Applications & future trends:

- **Applications** — Market Basket Analysis, fraud detection, customer segmentation, disease diagnosis, recommendation systems, weather forecasting.
- **Future trends** — Big Data and AI/Machine-Learning-based mining, cloud and real-time analysis, text and web mining, and the growing importance of privacy-preserving mining.

END OF EXPLAINED ANSWER PAPER

Unit – 2 : Data Warehousing and Data Mining — Prepared as per WBCHSE question pattern.